



DURBAN UNIVERSITY OF TECHNOLOGY
INYUVESI YASETHEKWINI YEZOBUCHWEPHESHE

FACULTY OF ACCOUNTING AND INFORMATICS
DEPARTMENT OF INFORMATION TECHNOLOGY

2023
MID YEAR SUPPLEMENTARY EXAMINATION

INSTRUCTIONAL PROGRAMME : Advanced Diploma in ICT

INSTRUCTIONAL OFFERING : Research Skills

PAPER NUMBER : 1

SUBJECT CODE/S : RESK401

DATE : 2023:07:21

DURATION : 2 HOURS

TIME : 9:00 - 11:00

TOTAL MARKS : 120

NUMBER OF PAGES : 20

EXAMINER/S : A Singh and L Manqele

MODERATOR/S : P Singh

INSTRUCTIONS/REQUIREMENTS:-

Answer all questions

Do not turn the page until permission is given

1. Which of the following statements correctly describes research?
 - i. A series of systematic activities undertaken to find a solution to a problem
 - ii. The process of designing and developing a software solution
 - iii. An intellectual inquiry or quest towards truth
 - iv. Aims to enhance existing knowledge
 - A. i, ii, iii and iv
 - B. i, ii, and iii
 - C. i, iii and iv
 - D. ii and iv

2. Given the list of stages below, which of the following is the correct sequence for conducting research?
 - i. Identifying the problem
 - ii. Determine the research strategy
 - iii. Data collection and analysis
 - iv. Review relevant literature
 - v. Draft research questions, objectives, and hypotheses
 - A. i, ii, iv, v, iii
 - B. i, v, iv, ii, iii
 - C. i, iii, iv, ii, v
 - D. i, iv, ii, iii, v

3. Which best defines the research process?
 - A. The researcher's plan of action to be followed when collecting data
 - B. A method of collecting research data
 - C. The steps the researcher follows in carrying out a research project
 - D. The account of a study the researcher will write for publication

4. Which activity should be done first?
 - A. Developing a research proposal
 - B. Formulating a research question
 - C. Deciding on the data analysis procedure
 - D. Performing data analysis

5. Pick the option that is not a characteristic of a good research problem
 - A. Requires a scientific process of enquiry
 - B. A researchable problem includes the collection and analysis of data
 - C. Multiple outcomes are possible
 - D. The solution is easily achieved

6. A clearly defined research problem helps the researcher to ____
 - i. Provide details of where the problem is occurring
 - ii. Formulate a research question and objectives
 - iii. Develop a conceptual framework
 - iv. Become familiar with the types of methodologies followed by other researchers
 - A. i, iii
 - B. i, ii
 - C. ii, iv
 - D. iii, iv

7. The literature review assists the researcher in ____
- A. Formulating a research question and objectives
 - B. Determining the research strategy
 - C. Understanding the state-of-the-art of an area
 - D. Writing the research proposal
8. What is the function of a literature review?
- A. Provide a theoretical background to the study
 - B. Identify a research problem
 - C. Provide reliable research findings
 - D. Demonstrate the use of online databases
9. A literature review helps to ____
- A. Bring clarity and focus to a research problem
 - B. Broaden the researcher's knowledge base in a research area
 - C. Understand the recent research methodologies
 - D. All options
10. Which of the following statements are correct?
- i. A set of research objectives are derived from the research question
 - ii. A hypothesis is a statement that cannot be proven to be true
 - iii. The hypothesis is developed, and then the research tests it
 - iv. The research question is derived from a set of research objectives
- A. i, ii
 - B. i, iii
 - C. ii, iii
 - D. ii, iv
11. Which of the following serves as a framework for the rest of the research study?
- A. Literature review
 - B. Research question
 - C. Research proposal
 - D. Problem identification
12. Which of the following does not describe research strategy?
- A. Determines how data will be collected and analysed
 - B. A plan containing methodological components to ensure that the objectives can be achieved
 - C. The guiding philosophy in formulating results
 - D. Makes a case for the need for this study
13. In data collection, which of the following best describes a primary source?
- A. data that is collected for the first time
 - B. online databases
 - C. data in the form of reports
 - D. data that has already been analysed
14. Which is used more often in quantitative analysis?
- A. interviews
 - B. closed questions
 - C. observations
 - D. focus-groups

15. Which of the following least applies to sampling?
- A. Selecting a small number of items from a much larger group
 - B. The entire population is included in the research
 - C. the research results are generalised to the population
 - D. the sample is an accurate representation of the whole population
16. Which two statements best describe data analysis?
- i. A process of careful examination of collected data, searching for patterns
 - ii. The task of selecting a small number of items for the research from a much larger group
 - iii. Conversion of textual data to numerical data via coding
 - iv. It is the numerical testing of data
- A. i and iii
 - B. i and ii
 - C. ii and iv
 - D. iii and iv
17. Which of the following skills are needed to write a good research proposal?
- i. Quantitative skills
 - ii. Scientific writing and grasp of nomenclature relevant to the field
 - iii. Search engine skills
 - iv. Critical thinking
- A. ii, iii
 - B. i, iv
 - C. ii, iv
 - D. i, iii

Answer questions 18 to 28 using Paragraph 1 below:

Paragraph 1:

Financial fraud is an ever growing menace with far consequences in the financial industry. Data mining had played an imperative role in the detection of credit card fraud in online transactions. Credit card fraud detection, which is a data mining problem, becomes challenging due to two major reasons – first, the profiles of normal and fraudulent behaviours change constantly and secondly, credit card fraud data sets are highly skewed. The performance of fraud detection in credit card transactions is greatly affected by the sampling approach on dataset, selection of variables and detection technique(s) used. This paper investigates the performance of naïve bayes, k-nearest neighbor and logistic regression on highly skewed credit card fraud data. Dataset of credit card transactions is sourced from European cardholders containing 284,807 transactions. A hybrid technique of under-sampling and oversampling is carried out on the skewed data. The three techniques are applied on the raw and preprocessed data. The work is implemented in Python. The performance of the techniques is evaluated based on accuracy, sensitivity, specificity, precision, Matthews correlation coefficient and balanced classification rate. The results shows of optimal accuracy for naïve bayes, k-nearest neighbor and logistic regression classifiers are 97.92%, 97.69% and 54.86% respectively. The comparative results show that k-nearest neighbour performs better than naïve bayes and logistic regression techniques.

18.Paragraph 1 is best described as a(an):

- A. Introduction
- B. Problem statement
- C. Abstract
- D. Conclusion

19.What is the aim of this research:

- A. To determine if ML can accurately detect credit card fraud
- B. To conduct ML experiments to assist the financial industry with credit card fraud detection
- C. To determine the best performing ML technique for credit card fraud detection
- D. To help the financial industries around the globe with credit card fraud detection

20.What is the problem of concern?

- A. The changing profile of credit card normal and fraudulent behaviour and skewed datasets
- B. Identifying which is the best performing ML technique for credit card fraud detection
- C. To predict credit card fraud using ML techniques
- D. Financial industries are at risk of huge financial losses

21.What is the most likely potential future research following this study?

- A. The integration of the best performing ML technique/s into an decision support tool for fraud detection
- B. Including more ML techniques in the experiment
- C. Extending the experiment to include more datasets
- D. To collect and analyse the opinions of financial industry practitioners

22.Which best classifies the research design used in this research?

- A. Case study
- B. Quantitative-experimental
- C. Survey
- D. Applied design science

23.Choose the most suitable main research question.

- A. How many ML techniques are needed to accurately detect credit card fraud?
- B. Which is the best performing ML technique for credit card fraud detection?
- C. How can credit card fraud be detected using ML techniques?
- D. Can ML models be used to detect credit card fraud?

24.What is the major finding of this study?

- A. naïve bayes, k-nearest neighbor and logistic regression performed equally well
- B. K-nearest neighbor is the best ML technique for this problem
- C. Under-sampling and oversampling are effective methods for skewed dataset
- D. Credit card fraud detection is possible with ML techniques

25. Which is the most plausible shortcoming of this research?
- ML techniques need to be integrated into real-life financial institutions to more precisely measure their effectiveness.
 - K-nearest neighbour is an outdated ML technique and hyper-parameter tuning would have improved results.
 - Only three ML techniques were used in the experiment.
 - The research was conducted using Python.
26. Does this research require research ethics approval?
- Yes
 - No
27. How was the data collected in order to conduct the experiment? Pick the most correct answer.
- It is not clearly stated
 - An online dataset/s
 - Financial institutions database
 - From banking staff
28. In this study, a comparative analysis was done on what?
- Three ML techniques
 - The data from European cardholders
 - Features of imbalanced data and skewed dataset
 - Over-sampling and undersampling methods
- **
29. A quality literature study is key to avoid ____
- repeating research that has successfully solved the problem
 - sources of information that are not recent and hence, no longer relevant
 - reinventing the wheel
 - all options
30. Which is the most reliable source of information for your literature review?
- TV documentary
 - Newspaper article
 - A peer-reviewed research article
 - An e-book
31. The secondary source of information comprises of ____
- Text books and research monographs
 - Subject periodicals and encyclopaedias
 - A summary of publications
 - Bibliography and patents
32. Articles published in research journals are called ____
- Reference sources
 - Secondary sources
 - Primary sources
 - Tertiary sources
33. Which one of the following is likely to be peer-reviewed?
- Lecture notes
 - Journal article
 - Poster
 - Textbook

34. Why is a primary literature source more valuable to a researcher than a secondary literature source?
- A. It reviews relevant literature
 - B. It gives general information about a particular topic
 - C. It contributes new knowledge from research
 - D. It synthesises the results from several related studies
35. Which of the following is one of the characteristics of popular sources of information?
- A. Authors include academic affiliations and credentials
 - B. Use of field-specific vocabulary or nomenclature
 - C. Little to no citations to primary sources of information
 - D. Peer reviewer
36. Having experts in the field review the validity of the information before it is published is called ____
- A. Journal review
 - B. Peer review
 - C. Colleague review
 - D. Information review
37. Which of the following sources of information may not have some degree of peer review?
- A. Journal articles
 - B. Conference proceedings
 - C. Thesis
 - D. Online magazines
38. The following is not a good gauge for the relevancy of the source?
- A. Abstract
 - B. Number of pages
 - C. Number of citations
 - D. Impact factor of the journal
39. Research strategy is best defined as ____
- A. A step-by-step plan to conduct research
 - B. It is part of the literature review
 - C. The process of creating a research proposal
 - D. It is the final stage of a research process
40. Which statement is false regarding quantitative research?
- A. Requires many respondents
 - B. Uses closed-ended questions
 - C. Focuses on testing theories and hypotheses
 - D. Mainly expressed in words
41. Which of the following statements about qualitative data is not accurate?
- A. Qualitative data is often collected using informal group discussions
 - B. An example of qualitative data is "75% of developers use C#"
 - C. The quality of qualitative data depends to a large extent on the way it is interpreted
 - D. The quality of qualitative data can be affected if one person dominates the group discussion

42. Which of the following variables cannot be expressed in quantitative terms?
- A. Ethnic group
 - B. Marital status
 - C. Economic status
 - D. Professional attitude
43. Which of the following statements about quantitative data is accurate?
- A. Quantitative data is easy to analyse
 - B. Quantitative data is collected using in-depth interviews
 - C. An example of quantitative data is "80% of IT students utilise Visual Studio"
 - D. Quantitative data is collected to find out the reasons for customers' behaviour
44. Qualitative data is mostly collected from?
- A. Interviews
 - B. Experiments
 - C. Surveys
 - D. ITS reports of student assessment marks
45. Quantitative data is mostly collected from?
- A. Open-ended survey
 - B. Case studies
 - C. Observation
 - D. Experiments
46. Which is a poorly formulated qualitative question?
- A. How satisfied are you with provisioned IT labs?
 - B. On a scale from 1-5, how satisfied are you with provisioned computer laboratories?
 - C. What is the most positive aspect of the IT labs
 - D. What can be done to improve the IT labs
47. When is it most appropriate to use a quantitative research approach?
- A. when confirming or testing a theory or hypothesis
 - B. when trying to understand a concept
 - C. when trying to comprehend experiences
 - D. when using open-ended questions
48. When is it most appropriate to use a qualitative research approach?
- A. when confirming a theory
 - B. when testing a hypothesis
 - C. when you want to understand a concept or experiences
 - D. when there are many respondents
49. Which research strategy would be most appropriate for the development or improvement of an AI algorithm?
- A. Case Study
 - B. Design research
 - C. A survey
 - D. None of these options

50. Which of the following is false about a Survey?

- A. A survey is intended to collect data from a large sample of participants
- B. The literature review is, in essence, a type of survey
- C. A survey is a process of designing and distributing questionnaires to participants
- D. A survey can only be done via an online questionnaire

51. How best would you describe a longitudinal survey?

- A. Used to collect data relevant at a point in time
- B. Involves the collection of data over a long period
- C. Allows for the researcher to determine the state of affairs at that moment in time
- D. Short and focused on one aspect of research

52. Which of the following is not a type of longitudinal survey?

- A. Census
- B. Trend
- C. Panel
- D. Cohort

53. A longitudinal survey in which data is collected from the same sample over time is called a _____

- A. Public opinion poll
- B. Trend survey
- C. Cohort survey
- D. Panel survey

54. A longitudinal survey which shows how participants' opinions change over time is called a _____

- A. Public opinion poll
- B. Trend survey
- C. Cohort survey
- D. Panel survey

55. What is one of the disadvantages of using surveys?

- A. The type of responses can be limited by the design or template used in the questionnaire
- B. A wide reach is achievable
- C. Online surveys are quick to develop and administer
- D. The analysis of collected data is faster due to statistical analysis tools

56. Which of the following is false about Design Science research?

- A. Includes the analysis of the performance of the new artefact through its use
- B. Aims to create new knowledge through the design of novel artefacts
- C. Relies completely on the collection and analysis of data
- D. Aims to improve on a current societal problematic situation

57. Which of the following are the correct stages of Design Research strategy?

- A. Awareness of the problem, suggestion, development, evaluation and conclusion
- B. Initiation, analysis, design, development and testing
- C. Planning, evaluation, development, testing and implementation
- D. Planning, evaluation, monitoring, testing and implementation

58. Which of the following best describes the Evaluation stage of the Design Research strategy?
- A. Envisioning plans for improvement which can include new elements
 - B. Reading of relevant literature and interaction with stakeholders
 - C. Packaging of the new design for patent or marketing
 - D. Artefacts are performance tested using analytical tools
59. Which of the following best describes the Awareness of the Problem stage of the Design Research strategy?
- A. Envisioning plans for improvement which can include new elements
 - B. Reading of relevant literature and interaction with stakeholders
 - C. Packaging of the new design for patent or marketing
 - D. Artefacts are performance-tested using analytical tools
60. An Experiment is best described as ____
- A. designing and distributing questionnaires to participants
 - B. collecting data from a large sample of participants
 - C. determining the relationship between the dependent variable and the independent variable
 - D. done to achieve a real-world goal
61. What is a hypothesis?
- A. A variable that must remain constant throughout the research process
 - B. It involves a systematic literature review
 - C. A proposed explanation for a phenomenon
 - D. The process of setting up an experiment to measure dependent variables
62. Hypothesis testing is the process of ____
- A. Formulating true and false hypotheses, which are then tested through experimentation
 - B. Collection and analysis of data and the results will either support or reject the hypothesis
 - C. Ensuring that controlled groups have exactly the same intervention as experimental groups
 - D. Conducting qualitative research
63. Which of the following are valid variables in an experiment?
- A. Independent variable
 - B. Dependent variable
 - C. Controlled variable
 - D. All options
64. Which statement best describes a controlled variable?
- A. Must remain constant throughout the process
 - B. Changed or manipulated by the researcher
 - C. May easily be adversely affected by other factors
 - D. Is the centre of attention during an experiment
65. Which of the following is a poor example of a hypothesis?
- A. Agile programmers are more productive in open-plan offices
 - B. Which UX design is best suited for mobile apps
 - C. A convolutional neural network is the most accurate algorithm for visual analysis of crop disease

- D. Internet users prefer Google to other search engines for school homework
66. Which of the following is not a type of experiment?
- A. Pre
 - B. Quasi
 - C. Exploratory
 - D. True
67. Which of the following is not a type of Case study?
- A. Exploratory
 - B. Descriptive
 - C. Explanatory
 - D. Confirmatory
68. Select the independent variable in the following hypothesis: "IT students will perform better in programming tests when using Python"
- A. IT students
 - B. Python
 - C. Test marks
 - D. Programming tests
69. Select the dependent variable in the following hypothesis: "IT students will perform better in programming tests when using Python"
- A. IT students
 - B. Python
 - C. Test marks
 - D. Programming tests
70. Which is not a good approach to conducting an experiment for the following hypothesis?: "IT students will perform better in programming tests when using Python"
- A. Perform pre-testing and post-testing of the same group of IT students
 - B. Divide the IT students into two groups; one group will use Python, and the other will use other languages
 - C. Perform post-testing of the same group of IT students
 - D. Divide the IT students into two groups; one group will use Python, and the other will use other languages; there will be pre-testing and post-testing of both groups
71. When conducting an experiment, the dependent variable is ____ while the independent variable is ____
- A. Manipulated; measured
 - B. Controlled; measured
 - C. Measured; manipulated
 - D. Controlled; manipulated
72. The researcher wants to survey software developers who have experience with a certain developer platform. She sets inclusion criteria to at least 6 months of experience and disregards those interested in using and training to use the platform. Choose the most correct sampling error.
- A. Sample frame error
 - B. Selection error
 - C. Confidence level error
 - D. Population-specific error

73. Which of the following is an example of purposive sampling?
- A. A sample of PhD students working in the area of Machine Learning
 - B. When the researcher does not consider whether the sample accurately represents the population
 - C. People who have many friends are more likely to be recruited into the sample
 - D. Where a research team is seeking opinions about gaming consoles among various age groups
74. When the sample size is increased, the impact on sampling is ____
- A. The confidence level will increase
 - B. There will be no impact
 - C. The amount of sampling error will be reduced
 - D. The sample mean will be closer to the population mean
75. Which statement is false?
- A. Increasing the sample size will reduce the size of the sampling error
 - B. There is no way to prevent sampling error except a census of the entire population
 - C. Sampling error will occur because the population mean is different to the sample mean
 - D. Sampling error is a normal occurrence, and researchers need to declare it as a z-value
76. Which of the following is most likely an example of a non-sampling error?
- A. Two researchers sampled students to find their preferred social media platform. They use the same methods and randomly select a sample. The first researcher finds that 45% prefer WhattheHeck, but the second finds that 51% prefer WhattheHeck.
 - B. Two researchers setup an online survey to collect responses in relation to the usability features of WhattheHeck. The survey can not be loaded on iPhones and iPads.
 - C. A researcher developed an app for students to record responses for their preferred SRC candidate before the election. One of the candidates' names is missing from the app.
 - D. A researcher interviews students to determine their level of satisfaction with library services. He uses pure random sampling.
77. An interview was conducted with highly satisfied users with a new software system feature. What type of sample error is this?
- A. Non-response error
 - B. Selection error
 - C. Convenience error
 - D. Population-specific error
78. What is the most likely initiative a researcher can use to mitigate the effects of non-response sampling error?
- A. Introduce an incentive
 - B. Stratify the population
 - C. Ensure the sample mean matches the population mean
 - D. Improve the quality of the questionnaire

79. Which is the most likely example of a sample frame error?
- A. A study was conducted to determine which programming language developers prefer. Software developers from Microsoft were used as the sample.
 - B. A software company conducts research on the usability of a newly installed delivery and deployment platform. All developers who used the platform were used as the sample.
 - C. A University wants to determine staff preference for online collaboration tools. 10 staff members were randomly selected and asked about their preferences.
 - D. A lecturer wants to determine the quality of their teaching materials. All students in her class were asked to complete a survey.
80. What sampling error is most likely to occur when using the snowball sampling technique?
- A. Non-response error
 - B. Selection error
 - C. Convenience error
 - D. Population-specific error
81. "Place the following IT student skills in order of importance: Maths; critical thinking; programming; team player. Which best describes this type of question when collecting data?
- A. Multiple choice
 - B. Closed
 - C. Ordinal scale
 - D. Rating scale
82. Which data collection method is best suited to a smaller number of participants; "how" and "why" type questions and seeking in-depth responses?
- A. Case study
 - B. Focus group
 - C. Observation
 - D. Survey
83. Which software would be best suited for creating an online questionnaire?
- A. EndNote
 - B. SurveyMonkey
 - C. Google sheets
 - D. Mendeley
84. A study aims to determine the software engineering project team factors that are considered most important for successfully implementing an online learning management system in a University. Which is the most appropriate data collection method for this study?
- A. Interview
 - B. Survey
 - C. Literature review
 - D. Case study

85. A study aims to determine students' perspective on the UI design of a university's newly implemented online learning management system. Which is the most appropriate data collection method for this study?
- A. Interview
 - B. Survey
 - C. Focus group
 - D. Online dataset
86. Which is the most likely research objective that will utilise an online dataset?
- A. To determine student perspectives
 - B. To compare the performance of different machine learning techniques
 - C. To generalise the findings of the sample to the population
 - D. To capture images of leaves for an image-processing experiment
87. Which is an example of a repository for online datasets?
- A. Google database
 - B. Microsoft Learn
 - C. Twitter
 - D. Kaggle
88. Good planning, number of participants, duration and location are factors that must be considered for the successful conduct of ____
- A. Survey
 - B. Interview
 - C. Focus group
 - D. Observation
89. A research objective is to involve client representatives in a sprint review session in order to collect feedback. Which is the best-suited configuration for the focus group?
- A. Two-way
 - B. Dual moderator
 - C. Client participation
 - D. Duelling moderator
90. A study intends to utilise a focus group to determine if ChatGPT should be used as a tutoring tool. Which is the best configuration for the focus group?
- A. Two-way
 - B. Dual moderator
 - C. Client participation
 - D. Duelling moderator
91. Which statement is not an advantage of using an online dataset?
- A. Avoids the time-consuming and costly activities of collecting data
 - B. Avoids transcribing data
 - C. Avoids tidying up data and ethics approval.
 - D. Cleaning of datasets is optional
92. What is the term for the statistical technique used to measure the strength and direction of a relationship between two variables?
- A. Analysis of variance
 - B. Regression analysis

- C. Correlation analysis
- D. Factor analysis

93. What is the term for identifying patterns and relationships in data?

- A. Data collection
- B. Data coding
- C. Data analysis
- D. Data visualisation

94. What is the term for the measure of the degree of association between two variables?

- A. Mean
- B. Median
- C. Mode
- D. Correlation

95. Null hypothesis refers to the fact that ____

- A. Nothing changed
- B. Something changed
- C. The results remained zero (0)
- D. The results were undefined

96. Alternative hypothesis refers to the fact that ____

- A. Nothing changed
- B. Something changed
- C. The results remained zero (0)
- D. The results were undefined

97. What is the term for the measure of the spread of data in a data set?

- A. Mean
- B. Median
- C. Mode
- D. Standard deviation

98. Which best describes data analysis?

- A. Pre-processing, transcribing and modelling data
- B. Converting images to digital format
- C. Downloading an online dataset; identifying and extracting factors
- D. All options

99. An automated warehouse management system utilises image recognition, barcode readers and IoT movement sensors. What type of data is this?

- A. Semi-structured
- B. Unstructured
- C. Structured
- D. Hybrid

100. A study aims to determine the most frequently used terms by IT students by analysing their chats in the virtual learning environment. Which is the most appropriate method for this?
- Transcribing
 - Descriptive statistics
 - Axial coding
 - Content analysis
101. What are the most correct sequence of steps for grounded theory methodology?
- transcribe data; arrange data; data coding; validate data
 - collect data; axial coding; open coding
 - arrange data; transcribe data; data coding
 - collect data; validate data; arrange data; selective coding
102. Which is a suitable tool for managing the qualitative data analysis process?
- EndNote
 - Mathlab
 - Nvivo
 - Stata
103. Which is a suitable tool for managing the quantitative data analysis process?
- Atlas-Ti
 - R
 - Nvivo
 - PowerHeda
104. A graph line's gradient (slope) can take on a value in the range -3; -2; -1; 0; 1; 2 and 3. Which best describes this type of data?
- Ratio
 - Interval
 - Ordinal
 - Nominal
105. The lecturer analyses the final marks to determine how many students scored distinctions. Which best describes this type of data?
- Ratio
 - Interval
 - Ordinal
 - Nominal
106. A questionnaire requires demographic details for race/ethnic group. Which best describes this type of data?
- Ratio
 - Interval
 - Ordinal
 - Nominal
107. A study aims to determine how many IT first-year students have done computing-related subjects in high school. Which best describes this type of data?
- Ratio
 - Interval
 - Ordinal
 - Nominal

108. Ethical issues are important in IT research because it ____
- A. helps the student pass the assignment
 - B. provides a guide for how researchers should interact with people
 - C. assists the researcher in writing up their findings
 - D. indicates that all people are very sensitive
109. Which best describes an ethics committee in a university? A group of academics who ____
- A. Are experts in their fields of research
 - B. Think about hypothetical research
 - C. Will consider a research proposal and the degree to which ethical issues have been catered for
 - D. Are useful to draw on when writing a proposal as a mentor
110. Which statement is FALSE regarding research ethics?
- A. Preserving the dignity, rights and welfare of research participants
 - B. Ensuring ethical research conduct and behaviour of the researcher
 - C. The truthfulness of data collection and subsequent results
 - D. Developing a research topic and a plan for the research undertaking
111. Why is it important that personal data about research participants are kept within secure, confidential records?
- A. So that the participants cannot find out what has been written about them
 - B. In case individuals, places, or organisations can be harmed through identification or disclosure of personal information
 - C. So that government officials and other people in authority can have easy access to the data
 - D. To enable the researcher to track down individuals and find out more about their lives
112. The two important components of IT research responsibility are sincerity in work and avoiding ____
- A. Plagiarism
 - B. Copyright
 - C. Peer review
 - D. Confidentiality
113. Which of the following is the most appropriate use of plagiarism detection software?
- A. Managing citations and producing a reference list
 - B. Comparison or similarity index to check for originality
 - C. Added features to help improve writing
 - D. To help the researcher in writing the literature review

114. Which of the following best describes fabrication of results?
- A. Changing or leaving out results in order to achieve the research objectives
 - B. Act of taking someone else's work or ideas and using them as your own
 - C. Act of insertion or modifying data when conducting fieldwork or experiments
 - D. It is a form of stealing
115. Which of the following best describes falsification of results?
- A. Act of taking someone else's word or ideas and using them as your own
 - B. Act of including or modifying data that never occurred when conducting fieldwork or experiments
 - C. It is a form of stealing
 - D. Changing or leaving out results in order to achieve the research objectives
116. Plagiarism detection software checks for plagiarism and reports this via ____
- A. Similarity index
 - B. Citation index
 - C. Content index
 - D. Word count
117. While conducting research that the DUT ethics committee has approved, two participants object to answering some questions which they considered to be too personal. As a result, they decide that they want to cease participating in the research. What are their responsibilities in this case?
- A. To complete the session because personal questions are important for the research
 - B. To explain to you why they wish to leave so that you can conduct a debriefing session
 - C. The participants have no responsibilities, and they can leave at any time
 - D. To report their grievances to DUT ethics committee
118. Informed consent means that researchers should tell participants about ____
- A. The hypothesis
 - B. Anything likely to affect the participant's decision to participate in the research
 - C. Details of any deception
 - D. Details of remuneration

119. Which statement is TRUE regarding research ethics?
- A. Relevant in experimental research
 - B. Relevant in design research
 - C. More relevant in experimental research than in survey research
 - D. Fabricating, falsifying or misrepresenting research data is not ethical behaviour
120. The process in which agreement to participate in research is obtained in writing and verified by signing a consent form is referred to as ____
- A. Informed consent
 - B. Revelation management
 - C. Expectation management
 - D. Participant decision-making

~end of questions total mark = 120~